Task 1:

IPv4 Address Classes & Real-World Use Cases

Objective: Classify IPv4 address ranges (Classes A through E) and map each class to a real-world enterprise or consumer application scenario.

IPv4 Classes Comparison Table:

IP Class	First Octet Range	Default Subnet Mask	Primary Purpose / Use Case	Real-World Example Device / Service
Class A	1.0.0.0 to 126.255.255.255	255.0.0.0 (/8)	Massive enterprise networks with millions of hosts.	**Large ISP Backbones / Apple Corporate Network** (e.g.,

IP Class	First Octet Range	Default Subnet Mask	Primary Purpose / Use Case	Real-World Example Device / Service
				17.0.0.0 network range).
Class B	128.0.0.0 to 191.255.255.255	255.255.0.0 (/16)	Medium to large size organizations and universities.	**College Campus Wi-Fi / Corporate Networks** (e.g., 172.16.0.0 internal network).
Class C	192.0.0.0 to 223.255.255.255	255.255.255.0 (/24)	Small local area networks (LANs) and	**Home Wi-Fi Router / Cafe Billing POS**

IP Class	First Octet Range	Default Subnet Mask	Primary Purpose / Use Case	Real-World Example Device / Service
			small businesses.	**Terminal** (e.g., 192.168.1.1 gateway).
Class D	224.0.0.0 to 239.255.255.255	N/A (Multicast)	Reserved for Multicasting data transmission.	**IPTV Live Streams / Zoom Video Conferencing / OSPF Routers.**
Class E	240.0.0.0 to 255.255.255.255	N/A (Experimental)	Reserved for experimental, research, and	**IEEE & IANA Internet Testing Protocols /**

IP Class	First Octet Range	Default Subnet Mask	Primary Purpose / Use Case	Real-World Example Device / Service
			future deployment.	**Military R&D Labs.**

Task 2:

Subnetting Calculation for 192.168.10.45 /24

Objective: Calculate the network address, broadcast address, and valid host IP range for a given IPv4 address and subnet mask with step-by-step working.

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1. Given Information:

- **Host IP Address:** 192.168.10.45
- **Subnet Mask:** 255.255.255.0 (CIDR Prefix: /24)

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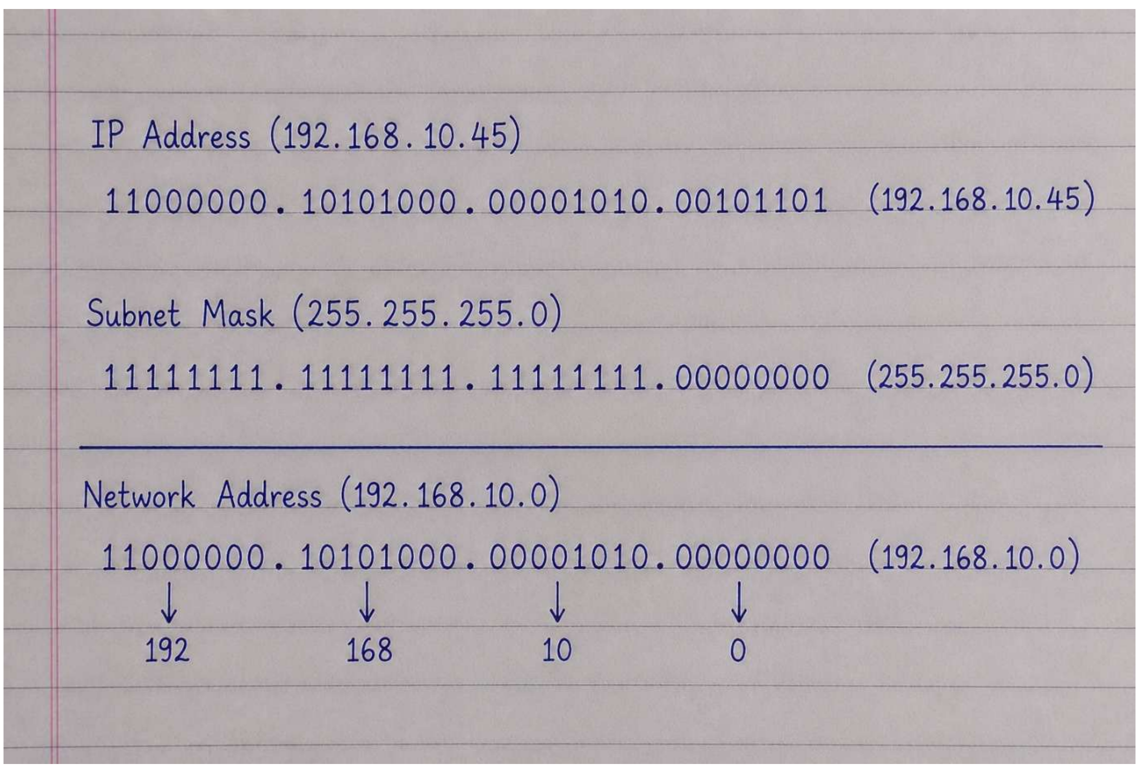
2. Step-by-Step Working & Calculation:

- **Step 1: Convert IP and Mask to Binary**
  - **IP Address (Binary):** 11000000.10101000.00001010.00101101
  - **Subnet Mask (Binary):** 11111111.11111111.11111111.00000000

- **Step 2: Calculate Network Address (Bitwise AND Operation)**
  - Performing bitwise AND operation between IP Address and Subnet Mask:



IP Address (192.168.10.45)

11000000. 10101000. 00001010. 00101101 (192.168.10.45)

Subnet Mask (255.255.255.0)

11111111. 11111111. 11111111. 00000000 (255.255.255.0)

Network Address (192.168.10.0)

11000000. 10101000. 00001010. 00000000 (192.168.10.0)

↓ 192 ↓ 168 ↓ 10 ↓ 0

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    - **Network Address: 192.168.10.0**

- **Step 3: Calculate Broadcast Address**
    - Set all 8 host bits (last octet) to 1s (11111111 = 255):
    - **Broadcast Address: 192.168.10.255**

- **Step 4: Determine Valid Host Range**
    - **First Usable Host IP:** Network Address + 1 = **192.168.10.1**
    - **Last Usable Host IP:** Broadcast Address - 1 = **192.168.10.254**
    - **Total Usable Hosts:** $2^8 - 2 = 256 - 2 = 254$ hosts

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3. Final Calculated Summary:

Parameter	Calculated Value
Network Address	192.168.10.0
Broadcast Address	192.168.10.255
First Usable Host IP	192.168.10.1
Last Usable Host IP	192.168.10.254
Valid Host IP Range	**192.168.10.1 to 192.168.10.254**

Task 3:

Subnetting Plan for Food Delivery Startup (Zomato Scenario)

Objective: Design a custom subnetting plan dividing the base network 10.0.0.0/24 into four equal subnets for Delivery Partners, Restaurants, Customers, and Admins.

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1. Subnetting Design & Calculation Details:

- **Base Network:** 10.0.0.0/24 (Default Subnet Mask: 255.255.255.0)
- **Subnets Needed:** 4 Subnets
- **Borrowed Bits:** 2 bits ($2^2 = 4$ Subnets)
- **New Subnet Mask:** /26 → **255.255.255.192**
- **Block Size (Step Size):** $256 - 192 = 64$ addresses per subnet
- **Usable Hosts per Subnet:** $64 - 2 = 62$ usable host IP addresses

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2. Complete Subnet Allocation Plan:

Subnet Group Name	Subnet Network Address	Subnet Mask (CIDR)	Valid Usable Host IP Range	Broadcast Address
Subnet 1: Delivery Partners	10.0.0.0	255.255.255.192 (/26)	10.0.0.1 to 10.0.0.62	10.0.0.63
Subnet 2: Restaurants	10.0.0.64	255.255.255.192 (/26)	10.0.0.65 to 10.0.0.126	10.0.0.127
Subnet 3: Customers	10.0.0.128	255.255.255.192 (/26)	10.0.0.129 to 10.0.0.190	10.0.0.191
Subnet 4: Admins	10.0.0.192	255.255.255.192 (/26)	10.0.0.193 to 10.0.0.254	10.0.0.255

Task 4:

College Wi-Fi Subnetting Calculation (172.16.0.0/16 for 20 Subnets)

Objective: Calculate the new subnet mask and usable host capacity per subnet to divide the 172.16.0.0/16 network into at least 20 building subnets.

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1. Step-by-Step Calculation Steps:

- **Step 1: Determine the Number of Borrowed Subnet Bits (n)**
  - We need at least 20 subnets ($2^n \geq 20$).
  - For $n = 4$: $2^4 = 16$ subnets (Insufficient)
  - For $n = 5$: $2^5 = 32$ subnets (Valid, satisfies the requirement)
  - **Borrowed Bits (n): 5 bits**

- **Step 2: Calculate the New Subnet Mask**
    - **Original CIDR:** /16 (Mask: 255.255.0.0)
    - **New CIDR Prefix:** /16 + 5 = /21
    - **Binary Subnet Mask:** 11111111.11111111.11111000.00000000
    - **Decimal Subnet Mask: 255.255.248.0**
- **Step 3: Calculate the Number of Available Hosts Per Subnet**
    - **Remaining Host Bits (h):** $32 - 21 = 11$ bits
    - **Total Addresses per Subnet:** $2^{11} = 2048$
    - **Usable Hosts Formula:** $2^h - 2$ (Subtracting Network & Broadcast IP)
    - **Usable Hosts per Subnet:** $2048 - 2 =$ **2046 hosts**

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2. Final Calculated Summary:

Parameter	Calculated Value
Base Network	172.16.0.0/16
Required Subnets	At least 20 (Creates 32 Subnets)
Borrowed Subnet Bits	5 bits
New Subnet Mask (Decimal)	255.255.248.0
New CIDR Notation	/21
Usable Hosts Per Subnet	2046 hosts

Task 5:

Explanation of Invalid IP Address (192.168.1.300)

Objective: Identify the configuration error in the provided IP address 192.168.1.300 and specify valid IPv4 formatting rules for the 192.168.1.0/24 subnet.

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1. **Explanation of the Error:**

- **Reason for Connection Failure:** The IP address **192.168.1.300 is invalid** and mathematically impossible in IPv4 networking.
- **Technical Cause:**
  1. IPv4 addresses consist of 4 octets separated by dots, where each octet is represented by 8 bits of binary data ($2^8 = 256$ total values).
  2. Because 8 bits can only represent values from **0 to 255**, no single octet in an IPv4 address can ever exceed **255**.

3. The fourth octet in 192.168.1.300 is **300**, which violates standard TCP/IP networking protocols, causing network interface cards (NICs) to reject the IP address.

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3. Correct & Valid IP Address Example:

For a standard 192.168.1.0/24 home or local subnet (Subnet Mask: 255.255.255.0):

- **Network Address:** 192.168.1.0 (Reserved)
- **Broadcast Address:** 192.168.1.255 (Reserved)
- **Valid Usable Host IP Range: 192.168.1.1 to 192.168.1.254**

Corrected Example:

Change the device IP to **192.168.1.30** or **192.168.1.50** with Subnet Mask 255.255.255.0 to restore network connectivity.